Energy Financial Risk Meter: Systemic Risk in the EU Energy Sector

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Outline

1. Introduction
2. Literature Review
3. Methodology
4. Data
5. EFRM Estimation & Structural Breaks
6. VAR Results
7. US Results & External Validity
8. Out-of-Sample Forecasting
9. Conclusion

Motivation

- Systemic risk monitoring became central after the 2008 GFC (Global Financial Crisis)
- Existing broad indicators (CISS, CLIFS, STLFSI2) aggregate across the entire financial system $\Rightarrow$ **sector-blind**
- The **energy sector** is unique:
  - Transmission channel for geopolitical shocks
  - Source of commodity-market volatility
  - Pillar of the real economy (inflation, fiscal, household welfare)
- 2021–2022 European energy crisis
  - TTF gas $\times 5$
  - spillovers to sovereign credit, industry, households
- **No dedicated systemic risk index exists for the energy sector**

Research Questions & Contribution

Research Questions

1. Can we build a real-time, sector-specific systemic risk indicator for EU energy?
2. How does geopolitical risk transmit to energy systemic risk, and has this channel changed?
3. Does EFRM carry out-of-sample predictive content beyond broad-based indicators?

Contribution

- First **Energy Financial Risk Meter** (EFRM): LASSO quantile regression on 100 EU energy firms + GPRD integration
- Two-stage time-varying transmission: GPRD $\rightarrow$ EFRM $\rightarrow$ equity / credit / US markets
- Out-of-sample horse race: EFRM vs. ECB's CISS

Literature & Gaps

Systemic Risk Measurement

- CoVaR (Adrian & Brunnermeier, 2016)
- MES / SRISK (Acharya et al., 2017)
- TENET (Härdle et al., 2016)
- FRM (Mihoci et al., 2020) $\rightarrow$ FRM@EM, FRM@RO, FRM@China

Energy Sector Risk

- EnSysRISK (Pierret, 2013)
- TENET on clean energy (Foglia et al., 2022)
- CoVaR + TENET on 100 global energy firms (Uddin et al., 2022)

Geopolitical Risk

- GPR Index (Caldara & Iacoviello, 2022)
- Wavelet GPR–energy connectedness (Ha et al., 2023)

Three Gaps

1. No FRM calibrated for the energy sector
2. Time-varying transmission to downstream financial markets uncharacterised
3. GPR not integrated into a unified network-based energy risk index

Methodology 1/3: EFRM Construction

Step 1: LASSO Quantile Regression ($\tau = 0.05$)
For each firm j, solve:

$$\min_{\alpha,\beta} \sum_{t=1}^{n} \rho_\tau\left(X^s_{j,t} - \alpha^s_j - A^{s\prime}_{j,t}\beta^s_j\right) + \lambda_s \|\beta^s_j\|_1$$

Step 2: Extract λ_j (penalisation parameter)
Higher $\lambda_j \Rightarrow$ more network connections needed to explain tail risk $\Rightarrow$ higher systemic exposure

Step 3: EFRM Index

$$\mathrm{EFRM}_t = \frac{1}{J}\sum_{j=1}^{J} \lambda_{j,t}$$

Monthly aggregation:

$$V_m = \frac{\sum_t \mathrm{EFRM}_t/\sigma_t}{\sum_t 1/\sigma_t}$$

Inverse-volatility weighting $\Rightarrow$ down-weights outlier days

Covariate vector $A_{j,t}$**:**

- Returns of 99 other firms
- TTF gas, WTI, Brent futures
- STOXX 600, STOXX O&G
- VSTOXX
- **Strictly European** (no US inputs)

Methodology 2/3: Rolling Window VAR

Reduced-form VAR(p) on rolling window of W observations:

$$y_t = c + A_1 y_{t-1} + \cdots + A_p y_{t-p} + u_t, \qquad u_t \sim \mathcal{N}(0, \Sigma_u)$$

Monthly specification:

- $y_t =$ (GPRD, EFRM, $\Delta\log$ TTF, $\Delta\log$ Stoxx)$'$
- $p = 2$ lags, $W = 48$ months
- Cholesky: GPRD $\rightarrow$ EFRM $\rightarrow$ TTF $\rightarrow$ Stoxx
- 12-month cumulative OIRF

Daily specification:

- $y_t =$ ($\Delta\log$ TTF, EFRM, $\Delta\log$ Brent, $\Delta\log$ Stoxx E)
- $p = 5$ lags, $W = 252$ days
- 21-day cumulative OIRF
- Ordering-invariance: $r > 0.998$

Stability screening: Companion matrix eigenvalues < 1; 63/65 monthly windows pass (96.9%).

Methodology 3/3: TVP-VAR (Kalman Filter)

State-space representation (Koop & Korobilis, 2014):

Observation equation:

$$y_t = X_t \beta_t + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \Sigma_t)$$

State transition:

$$\beta_t = \beta_{t-1} + v_t, \quad v_t \sim \mathcal{N}(0, Q_t)$$

Random walk on coefficients $\Rightarrow$ smooth, continuous parameter evolution

Forgetting factor:

$$\Sigma^{\beta}_{t|t-1} = \frac{1}{\lambda} \Sigma^{\beta}_{t-1|t-1}, \quad \lambda = 0.97$$

$\lambda = 0.97 \Rightarrow$ observations $\sim$33 periods old get negligible weight

Kalman update:

$$\beta_{t|t} = \beta_{t|t-1} + K_t(y_t - X_t \beta_{t|t-1})$$

Why both estimators?

Rolling VAR = hard window (locally constant). TVP-VAR = exponential discounting (continuously evolving). Agreement across both $\Rightarrow$ robust structural finding.

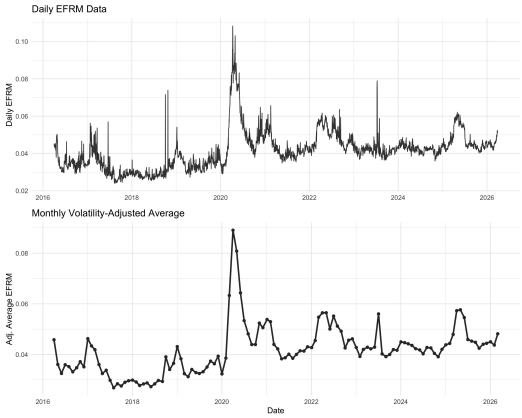
Data

Stage 1: EFRM Construction

- 100 European energy firms
- Daily stock returns + market cap
- $\tau = 0.05$ LASSO quantile regression
- April 2016 – March 2026

Stage 2: VAR Analysis

- *Monthly* ($N = 113$): EFRM, GPRD, CDS 5Y, TTF, Brent, Stoxx 600, Stoxx O&G
- *Daily* (∼2,350 days): TTF, Brent, Stoxx Energy, XLE, US T-bill
- Source: Refinitiv Eikon



Daily & Monthly EFRM Index.

EFRM Estimation & Structural Breaks

Bai–Perron structural break detection:

Monthly EFRM breaks:

- **February 2020** – COVID-19 onset
- **July 2021** – European gas-price run-up (TTF $\times 3$)

$\Rightarrow$ EFRM break is **7 months before** the invasion

Monthly GPRD break:

- **December 2021** – within ± 60 days of Feb 2022 invasion

Key Insight

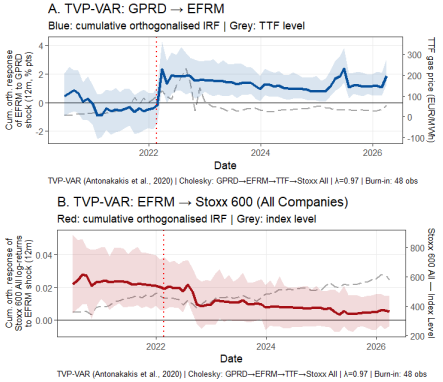
The Russia–Ukraine invasion **activated** the GPRD $\rightarrow$ EFRM transmission channel, but the EFRM regime shift had **already occurred** in July 2021.

$\Rightarrow$ EFRM has **early-warning** properties.

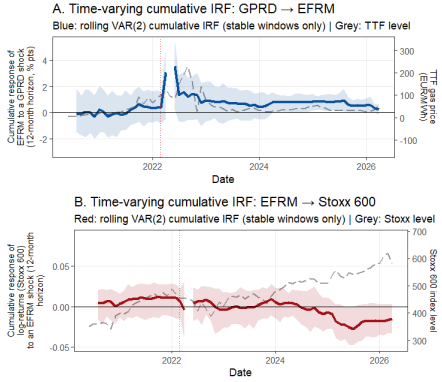
Stationarity:

- EFRM, GPRD: break-stationary
- TTF, Brent, CDS, equity: $I(1) \rightarrow$ log-returns

Stage 1: GPRD → EFRM — Channel Activation



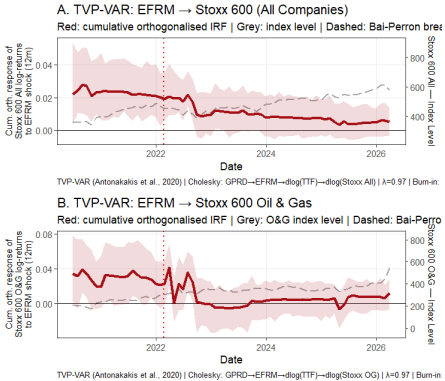
Rolling VAR: GPRD → EFRM



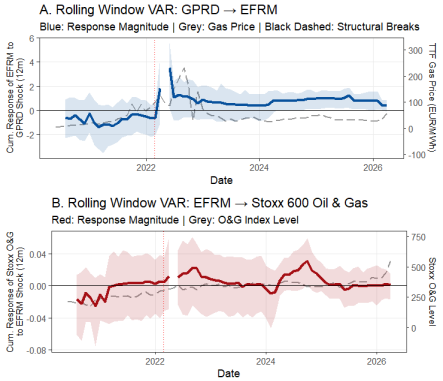
TVP-VAR: GPRD → EFRM (Panel A)

- Channel was **dormant** pre-2022; post-war amplification $\Delta = +1.52$ pp (TVP-VAR), confirmed by rolling VAR (+0.78 pp)
- End-plateau (+1.39 pp) **exceeds** the post-war mean — the channel may be *strengthening*

Stage 2: EFRM → European Equity Markets



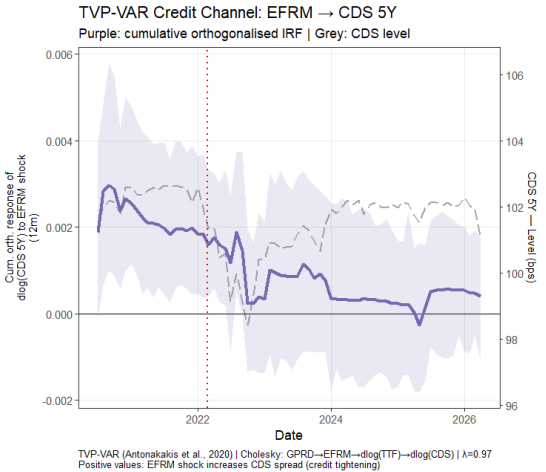
TVP-VAR: EFRM → Stoxx 600 All (A) & O&G (B)



Rolling VAR stress test

- Stoxx 600 All attenuates ≈60% (+0.023 → +0.009); Stoxx O&G attenuates 5-fold (+0.030 → +0.006)
- Both remain **positive** under TVP-VAR (no sign reversal); rolling VAR crosses zero marginally

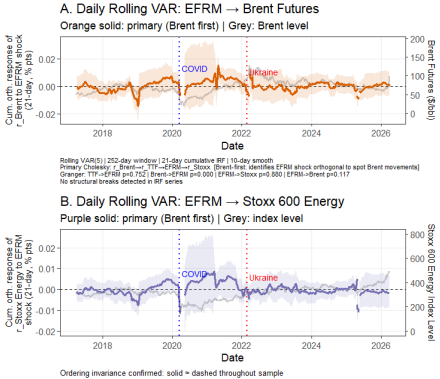
Credit Channel: EFRM → CDS 5Y

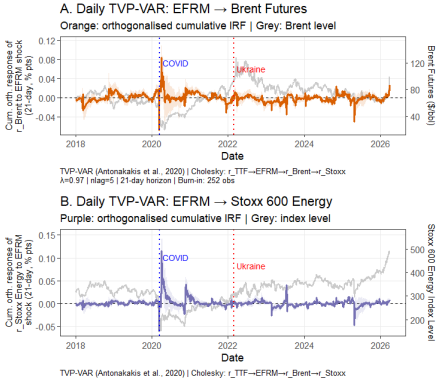


TVP-VAR: EFRM → CDS 5Y

- Credit channel attenuates ($+0.0022 \rightarrow +0.0006$) but remains **positive throughout** — unlike equity, which crosses zero
- **CDS informational superiority**: credit investors kept pricing energy default risk after equity normalised

Daily Cross-Frequency Validation (EU)



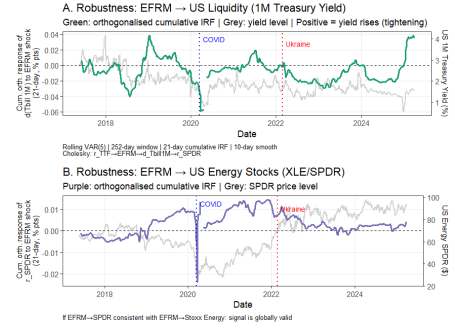


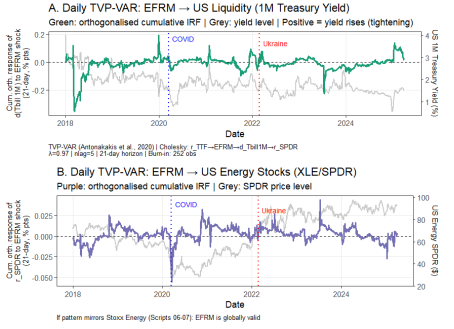
Rolling VAR: EFRM → Brent (A) & Stoxx E (B)

TVP-VAR: same channels

- **COVID** is the dominant activation (-0.060 to $+0.115$ pp V-shaped reversal); Ukraine weaker response
- Post-2023 normalisation to near zero confirmed by both estimators.

US External Validity: Treasury & Energy SPDR



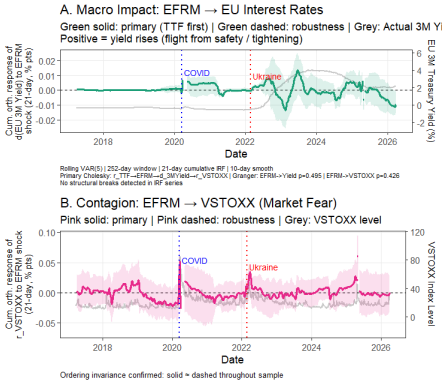


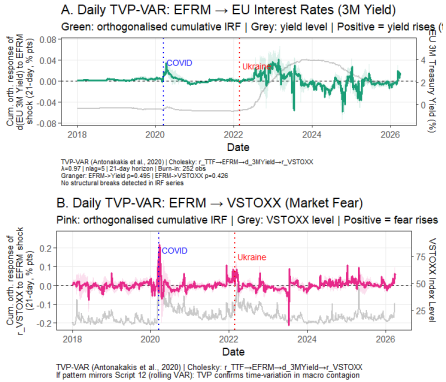
Rolling VAR: EFRM → US T-bill (A) & SPDR (B)

TVP-VAR: same channels

- **US Treasury**: persistently negative, intensifies at COVID. **US SPDR**: positive, Ukraine-centred peak (+0.044 pp)
- EFRM built from **EU-only covariates** ⇒ US transmission is a genuine external-validity test

EU Macro Financial Channel





Rolling VAR: EFRM → EU 3M Yield (A) & VSTOXX (B)

TVP-VAR: same channels

- **EU 3M Yield**: tightening at COVID. **VSTOXX**: large positive COVID & Ukraine peaks
- Always-present but **crisis-amplified** conduit — dual transmission (US + EU) confirms EFRM is a genuine systemic risk signal

Summary: All 11 Specifications

Channel	Pre-War	Post-War	Methods	Consist.?
GPRD → EFRM	Near zero	Strongly positive (+1.31 pp)	RV+TVP	Yes
EFRM → Stoxx All	Positive (+0.02)	≈60% attenuation	RV+TVP	Partial
EFRM → Stoxx O&G	Positive (+0.03)	5-fold attenuation	RV+TVP	Partial
EFRM → CDS 5Y	Positive (+0.002)	Positive near zero	TVP	N/A
EFRM → Brent (daily)	Near zero	COVID trough / Ukraine peak	RV+TVP	Yes
EFRM → Stoxx E (daily)	Near zero	COVID activation	RV+TVP	Yes
EFRM → US T-bill	Negative	Persistently negative	RV+TVP	Yes
EFRM → US SPDR	Near zero	Positive; Ukraine peak	RV+TVP	Yes
EFRM → EU 3M Yield	Near zero	Positive COVID peak	RV+TVP	Yes
EFRM → VSTOXX	Near zero	Large positive COVID peak	RV+TVP	Yes

Direction and timing consistent across **3 frequencies**, **2 estimators**, **2 continents**.

OOS Framework: EFRM vs. CISS Horse Race

Probit model with expanding-window estimation (no look-ahead):

$$\Pr(D_{t+h} = 1 \mid \mathcal{F}_t) = \Phi(\alpha + \boldsymbol{x}_t'\boldsymbol{\beta})$$

Model	Predictors	Rationale
M1: EFRM only	(EFRM_t)	Energy-sector tail interconnectedness
M2: CISS only	(CISS_t)	ECB broad-based benchmark
M3: EFRM+CISS	$(\mathrm{EFRM}_t, \mathrm{CISS}_t)$	Complementarity test
M4: Full	$(\mathrm{EFRM}_t, \Delta\mathrm{EFRM}_t, \mathrm{CISS}_t, \hat{\sigma}_t, r_t)$	Kitchen sink

Exercise 1: Daily tail events

- $D_{t+1} = \mathbb{I}(r_{t+1} < \hat{q}_{0.10,t})$
- Daily frequency
- $T_0 = 504$ days; 2,089 OOS; 229 events

Exercise 2: Vol exceedance

- $D^{\sigma}_{t+1} = \mathbb{I}(\hat{\sigma}_{t+1} > \hat{q}^{\sigma}_{0.75,t})$
- Monthly frequency
- $T_0 = 60$ months; 58 OOS; 11 events

OOS Results

Exercise	Model	T_{oos}	Events	AUC	95% CI	BSS
Ex. 1: Daily tail events ($\tau = 0.10$, $h = 1d$)						
	M4: Full	2,089	229	0.617	[0.576, 0.658]	+0.009
	M1: EFRM	2,089	229	**0.604**	[0.564, 0.644]	+0.014
	M2: CISS	2,089	229	0.598	[0.558, 0.638]	+0.006
	M3: EFRM+CISS	2,089	229	0.597	[0.556, 0.639]	+0.014
Ex. 2: Monthly vol exceedance ($\hat{\sigma} >$ 75th pctile)						
	M3: EFRM+CISS	58	11	**0.731**	[0.521, 0.941]	**+0.163**
	M1: EFRM	58	11	0.708	[0.480, 0.936]	+0.085
	M4: Full	58	11	0.685	[0.456, 0.913]	+0.090
	M2: CISS	58	11	0.673	[0.448, 0.898]	+0.028

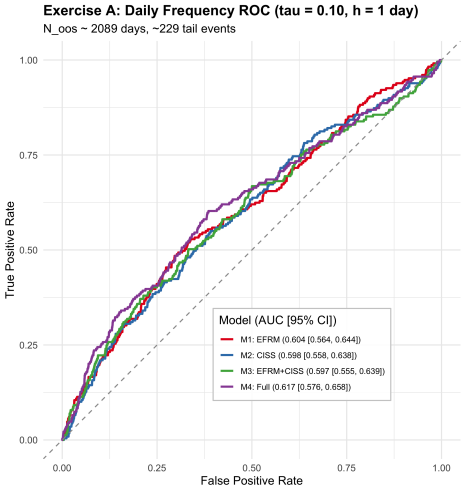
Headline Result

M3 (0.731) > M1 (0.708) > M2 (0.673)
EFRM alone > CISS alone; combined = **best**
BSS +16.3% ⇒ operationally useful

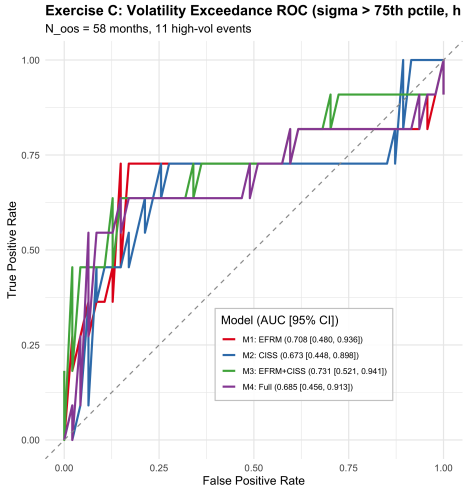
Interpretation

EFRM predicts **volatility regimes**, not crash timing — a *fragility* indicator. M4 overfits: AUC *drops* 0.731 → 0.685.

OOS: ROC Curves



Exercise 1: Daily — smooth curves (229 events)



Exercise 2: Vol exceedance — staircase (11 events)

Conclusion

What we built

The **EFRM**: Energy Financial Risk Monitor for daily/monthly energy sector risk monitoring.

What we found

1. Russia–Ukraine **permanently activated** GPRD → EFRM; may be *strengthening* by 2025
2. Equity transmission **attenuated** 60–80%; credit channel **persists** (CDS informational superiority)
3. **Global external validity**: US Treasury flight-to-safety + US SPDR positive — from EU-only inputs
4. **OOS**: EFRM + CISS achieves AUC 0.731 for volatility regimes (BSS +16.3%)

Policy implication

EFRM **complements** existing system risk indices (CISS) with sector-specific network information that broad indicators do not subsume.

Thank You

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